

Enhancing Garbage Classification with Swin Transformer and Attention-Based Autoencoder: An Efficient Approach for Waste Management

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Abstract. Annually, an enormous quantity of waste is generated, and the majority of it is discarded, resulting in environmental degradation. Employing computer vision-based techniques to classify waste into distinct recycling categories may provide efficiency in waste material management and disposal. Hence, this study proposes a novel lightweight approach that utilizes the Swin Transformer and attention-based autoencoder for garbage image classification which consumes significantly fewer resources. The proposed method achieves a high accuracy rate (94.27%) in identifying different types of waste materials on the TrashNet dataset. Our experimental results demonstrate that it outperforms several existing methods for waste classification with very limited parameters.

Keywords: Autoencoder · Classification · Self-supervised learning · Swin Transformer

1 Introduction

The growing emphasis on safeguarding the ecological environment and enhancing resource utilization has prompted worldwide initiatives to restructure waste classification. Population growth and economic development have led to a rapid increase in daily waste production. Improper management of waste can pose significant risks to the environment. One of the keys to addressing this issue is recycling. Since different items are recycled differently, proper classification of various garbage items is necessary. In numerous localities, individuals are engaged in the manual sorting of garbage at recycling plants or facilities. The aforementioned leads to a variety of diseases in the workforce (Sara et al., 2022).

In recent years, significant advancements have been made in automatic waste classification and detection. Large-scale image classification commonly relies on Convolutional Neural Networks (CNNs) as the prevailing architectures. However, one prominent drawback of CNNs is their restricted capability to consider the global context within an image. The introduction of the Vision Transformer (ViT) offered a solution to this problem. The idea is to divide the image into some small non-overlapping segments called patches and apply self-attention where every patch looks into every other patch to calculate the value of attention. The integration of the self-attention module in the ViT enables the model to incorporate information from the entire image, even at lower

layers. This global self-attention comes with an issue of computation. As each patch must consider every other patch, the computation is quadratic. The Swin Transformer was introduced as a solution to address this issue (Lilu et al., 2021). The concept is to divide the image into several windows, and whenever it computes self-attention, it looks into the patches of that particular window that provides local attention. Since the window is shifted, it allows the architecture to learn more about the context. Furthermore, the transformer also has a hierarchical fusion mechanism that combines features across different scales.

Convolutional autoencoders have emerged as a powerful tool in the field of image processing and analysis. The network's encoder compresses the input image into a lower-dimensional latent space, while the decoder reconstructs the original image from this compressed form. The architecture of the autoencoders is well suited for tasks such as image regularization and denoising, feature extraction, and even image generation (Yan et al., 2023).

Based on the aforementioned analysis, this research introduces a hybrid fusion model for automatic waste classification that combines the Swin transformer and autoencoder for garbage image classification. The performance of the autoencoder is boosted by integrating channel-wise attention in the encoder section to enhance feature representation. The experimental results exhibit a remarkable accuracy of 94.27% in the TrashNet dataset consisting of six classes. To summarize, the major contributions of this paper are as follows:

- A unique hybrid fusion model that combines a Swin Transformer and a channel attention-based autoencoder for accurate garbage image classification.
- The proposed model consumes significantly fewer resources in training compared to related state-of-the-art models.
- Through extensive experimentation, the proposed approach achieves a high accuracy rate in categorizing different types of wastes.

The article is organized as follows. Section 2 presents existing works for solving this problem succeeded by our solution in Section 3 and the results produced by our approach in Section 4. The paper concludes in Section 5.

2 Related Works

In recent years, researchers have extensively explored various deep learning based approaches to enhance the efficiency of garbage image classification tasks. Liu et al. (Liu et al., 2022). constructed a fusion model named GCNet that fuses two CNNs and a transformer. The CNN models that were chosen are Densenet and EfficientNetv2 and a transformer architecture, ViT was integrated. The proposed GCNet architecture produces a large number of parameters since it performs a stacked ensemble of large image classifiers. While Nnamoko et al. (Nnamoko et al., 2022) presented a lightweight 5-layer CNN-based architecture for solid waste image classification, this model failed to achieve a noteworthy classification performance in binary mode. Ma et al. (Ma et al., 2022) modified the ResNet-50 architecture by incorporating a convolutional block attention module (CBAM) that combines both channel and spatial attention

mechanisms. The modified ResNet-50 achieved an accuracy of 92.08% on the TrashNet dataset. Although Resnet-50 is a lightweight image classifier, the absence of transfer learning in the implementation increases the computational burden. Yuan et al. (Yuan et al., 2022) presented a two-stream-based architecture. In their approach, they split the dataset (with six classes) into two parts, each having three classes. In the first part of the architecture, the model classifies a garbage image into one of the two major categories, and the second part, constructed by ResNext, performs the classification based on the sub-division of the trash from the results of the first stream. Agarwal et al. (Agarwal et al., 2022) used two one-shot-based models named siamese network and triplet loss network for plastic waste classification. Among the two methods, the siamese network produced better performance. Qi et al. (Qi et al., 2023) employed the Mask R-CNN network with ConvNeXt as the backbone architecture to classify waste images on their custom dataset with 1660 manually labeled images. Malik et al. (Malik et al., 2022) utilized EfficientNet-B0 for categorizing garbage, and performed a comparative analysis of different versions of EfficientNet series. In the garbage classification process, however, the fine-tuned version of the EfficientNet-B0 architecture could not provide satisfactory performance. Fu et al. (Fu et al., 2021) used novel fine tuning in MobileNet by adding a convolution block before the model and some convolution blocks at the end of the MobileNet feature extractor. The architecture was found to outperform the base classifier by 6% reaching the accuracy of 92%. In another study, Shi et al. (Shi et al., 2020) introduced a novel multi-branch Xception network known as M-b Xception. The primary objective was to capture a greater amount of feature information at the expense of increased computation costs. The M-b Xception architecture achieved an accuracy of 94.3% on the TranshNet dataset by employing a substantial parameter count of approximately 20.8 million.

The majority of previous studies, however, either failed to achieve high performance in multiple evaluation metrics or consumed a significant amount of resources while training. In this paper, we present a lightweight hybrid approach that achieves a very high classification performance by training a comparatively small number of parameters.

3 Methodology

The proposed garbage image classifier construction process can mainly be divided into four parts. These are the data preprocessing, autoencoder construction, Swin Transformer fine-tuning, and finally, the full model construction. Section 3.1 holds a description of the dataset along with the preprocessing applied. Section 3.2 and 3.3 presents the autoencoder construction and Swin Transformer fine tuning respectively. Lastly, the full model construction process is described in Section 3.4. The Python implementation of this model can be found here¹.

¹<https://www.kaggle.com/code/naifislam/garbageclassificationwithselfsupervisedlearning97?scriptVersionId=124682361>

3.1 Dataset Description and Data Preprocessing

The dataset used in this experiment is TrashNet which is the most widely used benchmark dataset for garbage image classification (Mao et al., 2022). It consists of six classes (cardboard, glass, metal, paper, plastic, and trash) with a total of 2527 images of size 384×512. In the preprocessing stage, 20% of the data is first kept for testing and the rest of the data for training and validation. Since the dataset is slightly imbalanced, the subset kept for the training and validation data is balanced using random oversampling. The new dataset after data balancing is presented in Figure 1. The balanced data were randomly split into a ratio of 90:10 for training and validation. The images were resized to 224×224 to adjust with the input dimension of the pre-trained Swin transformer. To make the model more robust, real-time data augmentation techniques were used with random rotation, arbitrary height and weight shift, shear transformation, zooming, and flipping.

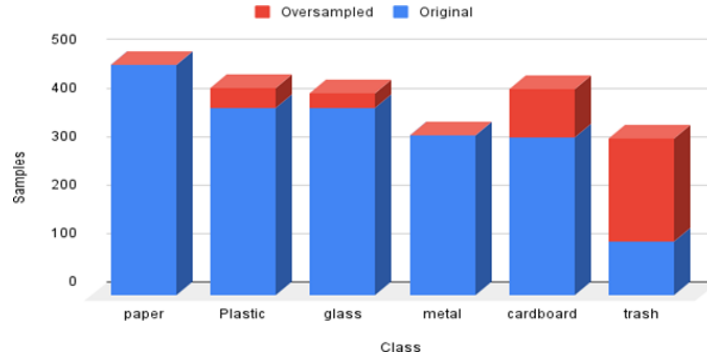


Figure 1. Balanced train and validation set of TrashNet dataset

3.2 Attention Based Autoencoder

Since the number of samples in the dataset is limited, traditional CNN architectures, even with transfer learning, tend to produce a low classification accuracy. In this scenario, self-supervised or unsupervised techniques like autoencoder can become a considerable option. Since features are compressed in the bottleneck, the encoder part can be used as a feature extractor. In this paper, we have constructed a convolutional autoencoder with channel attention at the encoder part to obtain higher performance. For channel attention, we have used Squeeze-and-Excitation (SE) block (Hu et al., 2018). In the squeeze process, the spatial dimensions of the feature maps are reduced for getting a summary of the channel-wise information. Formally, the statistic scalar z is generated by squeezing the input features U through its spatial dimensions $H \times W$, where the c^{th} element of z is calculated by Equation 1.

$$z_c = F_{eq}(u_c) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W u_c(i, j) \quad (1)$$

In the excitation stage, these scalars are passed through two fully connected (FC) layers. Let the weight matrix of FC layer 1 be denoted with $W1$ and FC layer 2 with $W2$. σ and

Relu are two activation functions applied after the layers. Therefore, the excitation process can be demonstrated as shown in Equation 2.

$$s = F_{ex}(z, W) = \sigma(W_2 \text{Relu}(W_1 z)) \quad (2)$$

Figure 2 presents a diagrammatic overview of the autoencoder. In the encoder part, 2×2 max-pooling was used for reducing the spatial dimension into half. Similarly, 2×2 upsampling was incorporated in the decoder for interpolation. The encoder's convolution layers comprise 48, 96, and 192 feature maps in three successive layers with a stride of 1. The bottleneck has 32 feature maps of 32×32 size. The autoencoder was trained on 50 epochs with Adamax as the optimizer and binary cross-entropy as the loss function. The number of parameters in the autoencoder is over 0.8 million.

3.3 Swin Transformer

Swin Transformer is a cutting-edge deep learning model based on transformers, which is renowned for its exceptional performance in computer vision-related tasks and outperforms most of the CNN and transformer models (Liu et al., 2021). The process of image classification with a Swin Transformer follows some number of steps. Firstly, the image is divided into fixed-size patches which are linearly projected into embedding vectors by multiplying with a learnable weight matrix. In the next phase, each patch is sequentially passed through a series of Swin Transformer stages which consists of shifted convolution and hierarchical self-attention.

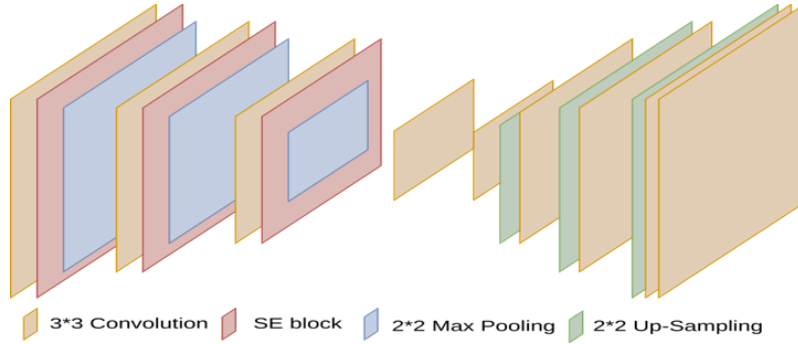


Figure 2. Attention Based Autoencoder

Finally, resulting hierarchical feature maps are processed through a classification head. In the proposed model, we integrated a pre-trained Swin Transformer for extracting features. These features are then passed through batch normalization and a multi-layer perceptron (MLP) before getting concatenated with the features extracted from the channel-attention-guided autoencoder.

3.4 Proposed Swin Transformer and Autoencoder-based Hybrid Fusion Model

The proposed model consists of two streams where one stream goes to the pre-trained Swin Transformer and the other stream to the autoencoder. In the Swin Transformer stream, all the parameters are frozen first to avoid unnecessary training. The features extracted from the Swin Transformer are then passed through a batch normalization layer to improve the training speed and generalization ability followed by an MLP block that consists of three fully connected layers. In the autoencoder stream, we take the encoder block of the autoencoder that was trained previously on the TrashNet dataset. The features obtained from the encoder are then passed through a SE block before further processing. The refined and recalibrated features from the SE block are then passed to the global average pooling layer (GAP). For transforming the 2D feature map into a 1D feature vector, the GAP layer was used instead of flattening. GAP not only increases the classification performance but also reduces the number of parameters by a significant margin. The feature vectors are then passed through a shallow MLP consisting of three feed-forward layers. After the small fine-tuning, the features from the Swin Transformer stream and the autoencoder stream are concatenated to construct the final feature vectors for classification. Figure 3 presents the architecture of the proposed model. The model was also trained on 50 epochs with the Adamax optimizer. Since the encoder part is already trained, the full model only requires the top layers to be trained with the labeled data. The overall trainable parameters are less than 1.4 million.

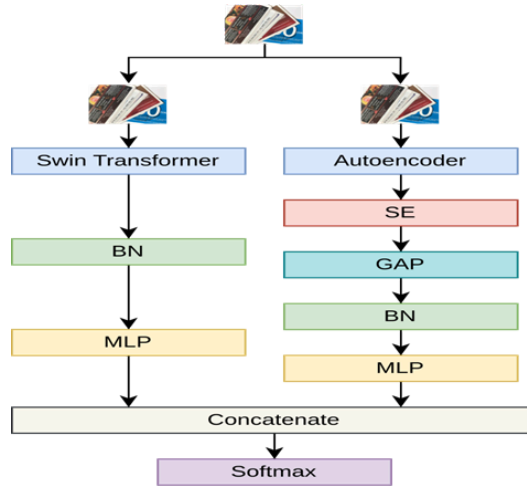


Figure 3. Proposed Model Architecture

4 Results

This section presents the performance of the proposed model along with a comparison with the previous studies.

4.1 Performance Evaluation

In this study, a number of critical measures, including accuracy, precision, recall, F1-score, and Matthews Correlation Coefficient (MCC), were used to evaluate the performance of the proposed model. For precision, recall, and F1-score, a macro average was taken.

Table 1. Summary of the performance

Sub-dataset	Accuracy	Precision	Recall	F1-score	MCC
Train	99.02	98.86	99.16	99.00	98.85
Validation	97.32	96.19	98.08	97.12	97.78
Test	94.27	93.07	92.88	92.89	92.96

Table 1 holds the performance analysis of the proposed model. The model has achieved a significant accuracy of 94.27% on the test set. In addition, Figure 4 showcases the confusion matrix for a detailed understanding of the model's performance. The results illustrate that the model is not biased toward a particular class. However, there remains a challenge of differentiating between paper and cardboard classes since all of the misclassified images of cardboard were predicted as paper. Furthermore, Figure 5 presents the receiver operating characteristic (ROC) curve. These results demonstrate that the model has strong discriminatory abilities.

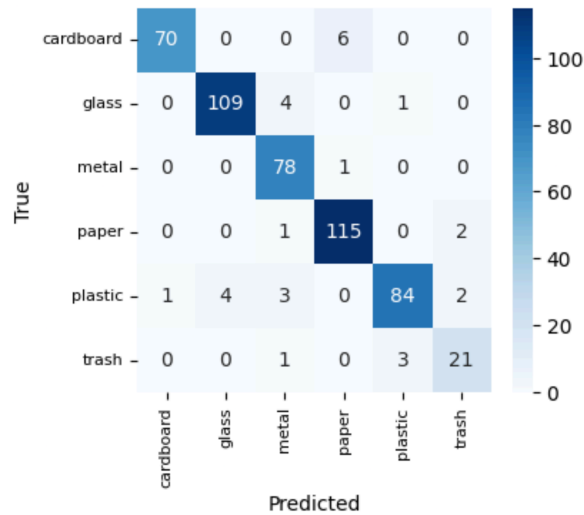


Figure 4. Confusion Matrix for the porpoised mode

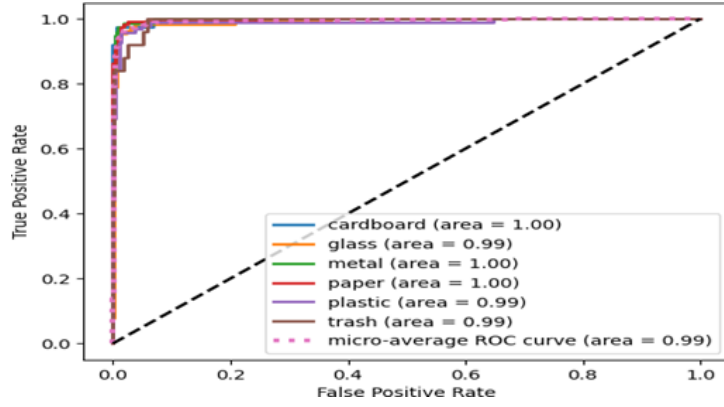


Figure 5. Receiver Operating Characteristic (ROC) curve for the porpoised model

Table 2. Comparison of performance between different models on TrashNet dataset.
- denotes not mentioned

Model	Trainable Parameters	Accuracy	Precision	Recall	F1-score
Attention Based ResNet-50 (Maur et al., 2023)	38M	92.08	-	-	-
Particle Swarm Optimization based late fusion (Ahmad et al., 2020)	350.5M	92.23	90.68	90.87	90.78
GoogleNet+MobieNetV2+ResNet-50 (Zheng et al., 2021)	-	92.85	-	-	88.25
Xception+attention based branch expansion (Zhang et al., 2021)	84.7M	91.80	-	-	-
Multi-brunched Xception (Shi et al., 2020)	20.8M	94.34	-	-	93.92
Swin Transformer + autoencoder (Proposed)	1.4M	94.27	93.07	92.88	92.89

4.2 Comparison with Existing Models on the TrashNet dataset

Since TrashNet is a small dataset and CNN models tend to perform better with a limited number of samples than the transformer-based computer vision models (Maur et al., 2023). For this, researchers tend to prefer CNN models when working with this dataset. A comparative analysis of different architectures is presented in Table 2. Ma et al. (Ma et al., 2022) incorporated channel and spatial attention mechanisms in the Resnet-50 architecture and achieved an accuracy of 92%. Several other popular CNN architectures were fine-tuned such as Xception with multi-branching (Zhang et al., 2021) (Shi et al., 2020). Authors also incorporated different ensemble architectures for enhancing the model such as combining features of CNN models (Ahmad et al., 2020), taking soft voting of three different pre-trained CNN models (Zheng et al., 2021). Most of the fusion-based models required a large number of parameters to be trained on.

5 Conclusion

This study addressed the need for precise garbage image categorization and provides a unique hybrid model that accomplishes this objective by effectively fusing a Swin Transformer and an attention-based autoencoder. The proposed architecture was assessed using the TrashNet dataset as a benchmark, yielding remarkable results with a classification accuracy of 94.27% while utilizing less than 1.4 million parameters for training. In comparison to similar studies, the proposed architecture showcased superior performance across various evaluation metrics. Furthermore, the study highlighted the model's ability to overcome challenges related to limited sample sizes. An oversight in this work is its focus on only six types of garbage, which might be different in real-world scenarios. In that case, the model might require small fine-tuning. Overall, the proposed method exhibited exceptional classification capabilities, positioning it as a valuable tool for precise garbage image categorization and smart waste management.

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